Structural Ruptures in Scientific Revolutions

A Formal Interpretation of Kuhnian Paradigm Shifts

Abstract

This paper proposes a formal interpretation of scientific revolutions within the structural framework of the Theory of Non-Derivability and Admissibility (TNA). The concept of paradigm introduced by Thomas Kuhn is reinterpreted as a structural admissibility operator S defining which theoretical configurations are meaningful within a scientific domain.

Normal science corresponds to internally derivable regimes N_0, where problem-solving operates through a dynamic $F : X \to X$. Scientific revolutions occur when structural admissibility changes, producing a rupture $S \to S'$.

Under this interpretation, paradigm shifts are not merely sociological transformations but structural transitions between distinct regimes of admissibility. The well-known phenomenon of paradigm incommensurability emerges naturally from the non-derivability of trajectories across structural rupture.

1 Introduction

In The Structure of Scientific Revolutions, Thomas Kuhn argued that scientific development does not proceed through linear accumulation of knowledge. Instead, it alternates between periods of stability and episodes of radical conceptual transformation.

During periods of “normal science,” researchers work within a shared conceptual framework or paradigm. This framework defines which questions are legitimate, which methods are acceptable, and which explanations count as valid.

However, the conceptual nature of paradigms has made them difficult to formalize. The present paper proposes a structural interpretation of paradigms using the TNA framework.

2 Paradigms as Structural Operators

Let

$$\Omega$$

denote the space of theoretical configurations.

A paradigm determines which configurations are admissible through a structural operator:

$$S : \Omega \to \{0, 1\}$$

where

- $S(\omega) = 1$ indicates theoretical admissibility
- $S(\omega) = 0$ indicates conceptual impossibility within the paradigm.

Thus, a paradigm is not merely a set of beliefs but a **constraint structure on intelligibility**.

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3 Normal Science as Internal Dynamics

Within a paradigm, scientific work proceeds through incremental theoretical refinement.

This process can be represented as a dynamic

$$F : X \to X$$

operating inside the admissible space

$$X = \{\omega \in \Omega \mid S(\omega) = 1\}.$$

This regime corresponds to what Kuhn called **normal science**.

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4 Anomalies and Structural Instability

Over time, empirical observations may produce anomalies.

In structural terms, anomalies correspond to configurations approaching the boundary

$$S(\omega) = 0.$$

When anomalies accumulate, the internal dynamic F becomes insufficient to maintain explanatory coherence.

This produces a crisis within the paradigm.

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5 Scientific Revolutions

A scientific revolution occurs when the admissibility operator changes.

Formally:

$$S \to S'$$

The new structure defines a new admissible space

$$X' = \{\omega \in \Omega \mid S'(\omega) = 1\}.$$

The previous theoretical dynamic cannot be extended across this transition.

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6 Incommensurability

The concept of paradigm incommensurability follows naturally.

If

$$S \neq S'$$

then

$$X \neq X'.$$

Therefore theoretical trajectories defined within X cannot be extended into X'.

This structural discontinuity explains why competing paradigms often appear mutually unintelligible.

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7 Examples

Examples of scientific structural rupture include:

PREVIOUS STRUCTURE	NEW STRUCTURE
Aristotelian physics	Newtonian mechanics
Newtonian mechanics	Relativistic physics
Classical physics	Quantum mechanics
Phlogiston theory	Oxygen chemistry

Each transition represents a change in admissibility conditions rather than a continuous derivation.

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8 Conclusion

The Kuhnian concept of paradigm can be interpreted as a structural operator defining admissibility within a scientific domain.

Scientific revolutions correspond to structural ruptures in which this operator changes.

This interpretation provides a formal explanation for the discontinuous character of scientific change and clarifies the origin of paradigm incommensurability.

Under this framework, scientific progress appears not as a smooth accumulation of knowledge but as a sequence of structurally constrained regimes separated by non-derivable transitions.